Syllabus of Engineering Economics & Project Management

Course Code:	OE302
Course Title:	Engineering Economics & Project Management
No. of Credits:	3 (L: 3, T: 0, P: 0)
Prerequisites:	NIL
Course Category:	Open Elective (Compulsory for all branches)

Course Objectives:

- To acquire knowledge of basic economics to facilitate the process of economic decision making.
- To acquire knowledge on basic financial management aspects.
- To develop the idea of project plan, from defining and confirming the project goals and objectives, identifying tasks and how goals will be achieved.
- To develop an understanding of key project management skills and strategies.

Group-A

Unit-I (INTRODUCTION, THEORY OF DEMAND & SUPPLY) [9 hours]

1.1 Introduction to Engineering Economics, the relationship between Engineering and Economics

1.2 Resources, scarcity of resources, and efficient utilization of resources.

1.3 Opportunity cost, Rational Choice Theory

1.4 **Theory of Demand:**

- The law of demand
- Different types of demand (Individual demand & Market demand)
- Determinants of demand
- Demand function
- Change in demand (Shift of demand curve) and the change in quantity demanded.
- Definition and types of Elasticity of demand (price, income & cross price elasticity) with mathematical derivation, Concept of elastic and inelastic goods, Measurement of price elasticity of demand (Point elasticity and Arc elasticity), Variation of price elasticity on different points of a linear demand curve, ranging from zero to infinity, Relationship between price, total revenue and price elasticity of demand (mathematical derivation).

1.5 **Theory of Supply**:

- Definition of supply
- Determinants of supply
- Supply function
- Supply curve and shift of supply curve.

1.6 **Market mechanism:**

- Definition of Market
- Price mechanism: determination of equilibrium price and quantity demand & supply (Numerical examples with graphical illustration).
- Stability of equilibrium.
- Basic comparative static analysis: Change in equilibrium due shift of demand & supply curve (Numerical problems with graphical illustration).

Unit-II (THEORY OF PRODUCTION & COSTS) [10 hours]

2.1: **Theory of Production:** Concept of production (goods & services), Different factors of production (fixed and variable factors), Short-run Production function (Graphical illustration), law of return (graphical and mathematical derivation), and Long run production function (returns to scale).

2.2: **Theory of Cost:** Short-run and long-run cost curves with graphical illustration, basic concept on total cost, fixed cost, variable cost, marginal cost, average cost etc. with the diagrammatic concept., Relationship between AC AND MC.

2.3: Economic concept of profit, profit maximization (numerical problems)

UNIT-III (DIFFERENT TYPES OF MARKET AND ROLE OF GOVERNMENT) [4 hours]

3.1: Perfect Competition: Features of Perfectly Competitive Market.

3.2: Imperfect Competition: Monopoly, Monopolistic Competition, and Oligopoly.

3.3: Role of government in Socialist, Capitalist and Mixed Economy structure with example.

<u>Group-B</u>

Unit-I (CONCEPT OF PROJECT) [4 hours]

1.1: Definition and classification of projects

1.2: Importance of Project Management.

1.3: Project life Cycle [Conceptualization→Planning→Execution→Termination]

Unit-II (FEASIBILITY ANALYSIS OF A PROJECT) [10 hours]

2.1: Economic and Market analysis.

2.2: Financial analysis: Basic techniques in capital budgeting – Payback period method, Net Present Value method, Internal Rate of Return method.

2.3: Environmental Impact study – adverse impact of the project on the environment.

2.4: Project risk and uncertainty: Technical, economical, socio-political, and environmental risks.

2.5: Evaluation of the financial health of a project – Understanding the basic concept of Fixed & Working Capital, Debt & Equity, Shares, Debentures etc., and different financial ratios like Liquidity Ratios, Activity Ratios, Debt-equity ratio & Profitability Ratio (Basic concept only).

N.B: Knowledge of financial statements is not required; for the estimation of ratios the values of the relevant variables will be provided.

Unit-III (PROJECT ADMINISTRATION) [8 hours]

3.1: **Gantt Chart** – a system of bar charts for scheduling and reporting the progress of a project (basic concept).
3.2: **Concept of Project Evaluation and Review Technique (PERT)** and **Critical Path method (CPM)**: basic concept and application with real-life examples.

Examination Scheme:

A. Semester Examination pattern of 60 marks:

1. Objective type Question (MCQ, Fill in the blanks, and Very Short question-1 mark each): At least five questions from each unit. [Total marks: 20]

2. Subjective questions: Five questions to be answered taking at least two from each group. [Total marks: 5x8=40]

B. Assignment (10 Marks)

Guideline for Assignment (10 Marks)

Students may be instructed to prepare a report on a project (preferably the based on the Major Project in 6th Semester), using a popular project management software in IT/Computer Laboratory, under the guidance of the Lecturer in Computer Science & Technology and Lecturer in Humanities.

C. Class Test: Two examinations 20 marks each. Take best of two.

D. Attendance: 10 Marks

Course Title :	CLOUD COMPUTING
Course Code	COPE307/2
Number of Credits :4	4 (L: 3, T: 1, P: 0)
Prerequisites	Netwoking Concepts
Course Category	PC
Course code : CST	Semester : SIXTH
Duration : 15 weeks	Maximum Marks : 100
Teaching Scheme	Examination Scheme
Theory : - 4 hrs/week	Continuous Internal Assessment : 20 Marks
Lectures:-3hrs/week Tutorial: 1 hr/week	Attendance-10 Marks
Total Contact Hours:60 Hours	Viva/Presentation/Assignment /Quiz etc : - 10 Marks
Practical : NIL	End Semester Examination : 60 Marks
Aim:	It will provide the students basic understanding about Cloud Computing, virtualization along with its security aspects and how one can migrate over it.

Course Objectives:

1. To learn the fundamental ideas behind Cloud Computing, the evolution of the paradigm, its applicability; benefits, as well as current and future challenges.
2. To understand the basics of cloud delivery models.
3. To learn about different virtualization techniques that serve in offering software, computation and storage services on the cloud.
4. To Analyze the Strategies for Secure Operation the cloud and list of the security requirements
5. To comprehend the basic ideas of different cloud tools and applications.

Course Content:

Contents (Theory)	Hrs	Marks	Module
UNIT 1: Cloud Computing Fundamentals	11	11	A
• Origins of Cloud computing. Fundamental concepts and models, Roles and boundaries. • Cloud components. • On-demand self-service, Broad network access, Location independent resource pooling, Rapid elasticity, Measured service. • Comparing cloud providers with traditional IT service providers, Roots of cloud computing • Migrating to clouds.			
UNIT 2: Cloud Delivery Model	11	11	A
❖ Cloud Delivery Models, The SPI Framework. ❖ Cloud Service Models. ❖ Cloud Deployment Models. ❖ Alternative Deployment models ❖ Expected benefits of the models.			

UNIT 3: Virtualization	12	12	B

- Characteristics & Taxonomy of virtualization.
- Virtualization vs Private Cloud.
- Desktop Virtualization, Hardware Virtual Machine (HVM).
- Virtual Servers.
- Logical Network Perimeter, Network Virtualization
- Data Center virtualization, Cloud Storage Device, Cloud usage monitor, Resource replication.

UNIT 4: Fundamental Cloud Security	14	14	B

- Cloud Information Security Objectives.
- Cloud Security Services & Relevant Cloud Security Design Principles
- Secure Cloud Software Requirements.
- Secure Development practices, Approaches to Cloud Software Requirement Engineering.
- Privacy and Compliance Risks, Threats to Infrastructure,
- Data and Access Control, Cloud Service Provider Risks.
- Cloud Security Policy Implementation.

UNIT 5: Cloud Tools and applications	12	12	C

- Cloud Performance Monitoring tools
- General model for Application platform
- Apache Virtual Computing Lab, VMWare, CloudSim.
- Microsoft Cloud Services (-Azure), Google Cloud Applications, Amazon cloud services.

Reference Books

1. "Cloud Computing Concepts, Technology & Architecture"- Thomas Erl, Zaigham Mahmood, and Ricardo Puttini , PrenticeHall
2. "Cloud computing a practical approach" - Anthony T.Velte , Toby J. Velte Robert Elsenpeter, TATA McGraw- Hill
3. "Cloud Computing (Principles and Paradigms)"- Rajkumar Buyya, James Broberg, Andrzej Goscinski, John, Wiley & Sons
4. "Cloud Computing"-Shailendra Singh, Oxford
5. "Cloud Computing-A Practical approach for learning and Implementation"-A Srinivasan & J. Suresh, Pearson

Course outcomes:

- *Analyze the Cloud computing setup with its vulnerabilities and applications using different architectures*
- *Apply and design suitable Virtualization concept, Cloud Resource Management*
- *Assess cloud Storage systems and Cloud security, the risks involved, its impact and develop cloud application*
- *Can understand the basics of security service models.*
- *Analyze the Strategies for Secure Operation the cloud architecture and list the security requirements.*

Unit No.	Unit Title	Group	Distribution of Theory Marks			
			R Level	U Level	A Level	Total
1.	Cloud Computing Fundamentals	A	4	4	3	11

West Bengal State Council of Technical & Vocational Education and Skill Development (Technical Education Division)

Course Title: Machine Learning	
Course Code	OE302
Number of Credits: 3 - L: 3, T: 0, P: 0	
Prerequisites	Concept of AI
Course Category	PC
Course code: CST	Semester: Sixth
Duration: 15 weeks	Maximum Marks: 100
Teaching Scheme	Examination Scheme
Theory: 3 hrs./week Total Contact Hours: 45 Hours	Continuous Internal Assessment: 20 Marks Attendance: 10 Marks Viva/Presentation/Assignment/Quiz etc.: 10 Marks End Semester Examination: 60 Marks
Aim of the Course	
This course will introduce the concept of Machine Learning through different learning methods.	

Course Objectives

- To learn the concept of how to learn patterns and concepts from data without being explicitly programmed
- To design and analyze various machine learning algorithms and techniques with a modern outlook focusing on recent advances.
- Explore supervised and unsupervised learning paradigms of machine learning.
- To explore Neural Network and Genetic Algorithm.

Course Content:

Contents (Theory)	Hrs./Unit	Marks
Unit 1: Supervised Learning (Regression & Classification)	15	20
❖ Basic methods: Distance-based methods, Nearest-Neighbours, Decision Trees, Naive Bayes ❖ Linear models: Linear Regression, Logistic Regression, Generalized Linear Models ❖ Introduction to Support Vector Machines, Nonlinearity and Kernel Methods		
Unit 2: Unsupervised Learning	7	10
• Clustering: K-means/Kernel K-means • Dimensionality Reduction: PCA and kernel PCA • Matrix Factorization and Matrix Completion		

UNIT 3: Artificial Neural Network	8	10
❖ Neural network representation ❖ Perception ❖ Multilayer Network and Back Propagation Algorithm ❖ Illustrative Example: Face recognition		
UNIT 4: Genetic Algorithm	8	10
❖ Representing Hypotheses ❖ Genetic Operators ❖ Fitness Function and Selection ❖ Hypothesis space search ❖ Genetic Programming		
UNIT 5: Reinforcement Learning	7	10
❖ **Introduction** ❖ **The Learning Task** ❖ **Q Learning** ❖ **Temporal Difference Learning**		
❖ **Note: Implementation of Machine Learning Algorithm by using suitable software can be done in Project work. Also seminar can be presented on topics of this subject.**		
Course outcomes		

Student should be able to

Sl. No.	Description	Bloom's Taxonomy Level
1	Understand the concept of machine learning.	**Knowledge, Understand**
2	Identify the regression and classification problem.	Analyze
3	Relate the supervised, unsupervised learning in the real life problem.	**Analyze**
4	Evaluate the machine learning models with respect to the performance parameters.	**Analyze**
5	Design and implement various machine learning algorithms in the range of real world problems.	**Design**

Reference Books:

Name of Authors	Title of the Book	Edition	Name of the publisher
Tom M. Mitchell	Machine Learning	-	Mc Graw Hill
Christopher Bishop	Pattern Recognition and Machine Learning	-	Springer
Rajiv Chopra	Machine Learning	-	Khanna Publishing House
Christopher M. Bishop	Pattern Recognition and Machine	-	Springer

Course Title	**Entrepreneurship and Start-ups**
Course Code	HS 302
Number of Credits	3
Pre Requisites	None
Total Contact Hours	3(L: 2; T: 1)/week = 45 hrs
Course Category	HS

Course Learning Objectives

1. To raise awareness, knowledge and understanding of enterprise/ entrepreneurship.
2. To motivate and inspire students toward an entrepreneurial career.
3. To understand venture creation process and to develop generic entrepreneurial competences.
4. To introduce students to the basic steps required for planning, starting and running a business.
5. To familiarise students with the different exit strategies available to entrepreneurs.

Course Outcomes:

After completing the course students will able to:

CO 1	Identify qualities of entrepreneurs, develop awareness about entrepreneurial skill and mindset and express knowledge about the suitable forms of ownership for small business
CO 2	Comprehend the basics of Business idea, Business plan, Feasibility Study report, Project Report and Project Proposal
CO 3	Understand the concept of start-up business and recognise its challenges within legal framework and compliance issues related to business.
CO 4	Make a Growth Plan and pitch it to all stakeholders and compare the various sources of funds available for start-up businesses

Detailed Course Content

Unit	Name of the Topic	Hours

1.	**ENTREPRENEURSHIP – INTRODUCTION AND PROCESS**
	• Concept, Competencies, Functions and Risks of entrepreneurship
	• Entrepreneurial Values& Attitudes and Skills
	• Mindset of an employee/manager and an entrepreneur
	• Types of Ownership for Small Businesses
	    o Sole proprietorship
	    o Partnerships
	    o Joint Stock company- public limited and private limited
		10

	companies • Difference between entrepreneur and Intrapreneur	
2.	**PREPARATION FOR ENTREPRENEURIAL VENTURES** • Business Idea- Concept, Characteristics of a Promising Business Idea, Uniqueness of the product or service and its competitive advantage over peers. • Feasibility Study – Concept – Locational, Economic, Technical and Environmental Feasibility. Structure and Contents of a standard Feasibility Study Report • Business Plan – Concept, rationale for developing a Business Plan, Structure and Contents of a typical Business Plan • Project Report- Concept, its features and components • Basic components of Financial Statements- Revenue, Expenses (Revenue & capital exp), Gross Profit, Net Profit, Asset, Liability, Cash Flow, working capital, Inventory. Funding Methods-Equity or Debt. Students are just expected to know about the features and key inclusions under, Business Plan and Project Report. <u>They **may not** be asked to prepare a Business Plan/ Project Report/ Project Feasibility Report in the End of Semester Examination.</u>	20
3.	**ESTABLISHING SMALL ENTERPRISES** • Legal Requirements and Compliances needed for establishing a New Unit- o NOC from Local body o Registration of business in DIC o Statutory license or clearance o Tax compliances	03
4.	**START-UP VENTURES** • Concept & Features • Mobilisation of resources by start-ups: Financial, Human, Intellectual and Physical • Problems and challenges faced by start-ups. • Start-up Ventures in India – Contemporary Success Stories and Case Studies to be discussed in the class. Case studies have been included in the syllabus to motivate and inspire students toward an entrepreneurial career from the success stories. <u>No questions are to be set from the case studies.</u>	04

5.	**FINANCING START-UP VENTURES IN INDIA** • Communication of Ideas to potential investors – Investor Pitch • Equity Funding, Debt funding – by Angel Investors, Venture Capital Funds, Bank loans to start-ups • Govt Initiatives including incubation centre to boost start-up ventures • MSME Registration for Start-ups –its benefits	06
6.	**EXIT STRATEGIES FOR ENTREPRENEURS** • Merger and acquisition exit, Initial Public Offering (IPO), Liquidation, Bankruptcy – **Basic Concept only**	02

Examination Scheme

❖ **End Semester Examination: 60 marks**

Suggested Question Paper Scheme for End Semester Examination

Group A: 20marks

Question Type	Number of questions to be set	Number of questions to be answered
MCQ, Fill in the blanks, True or False (Carrying 1 mark each)	25	20

Group B: 40marks

Question Type	Number of questions to be set	Number of questions to be answered
Subjective Type questions (Carrying 8 marks each)	10	5

❖ **Internal Assessment: 40 marks**
- Class test : 20 marks
- Assignment: 10 marks
- Class attendance: 10 marks